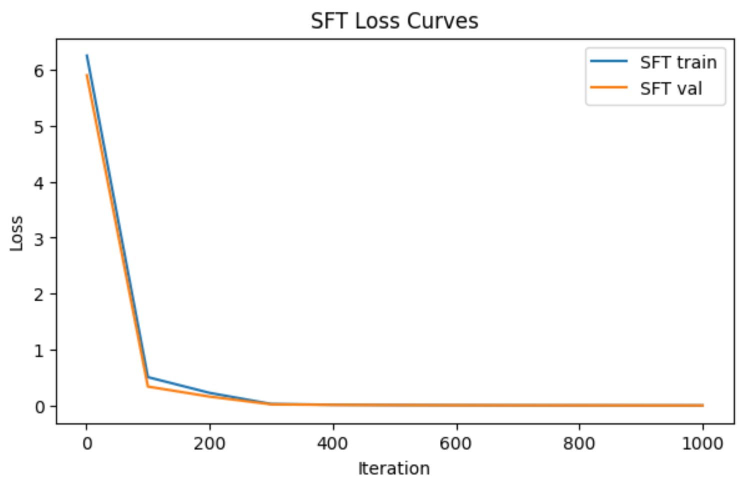
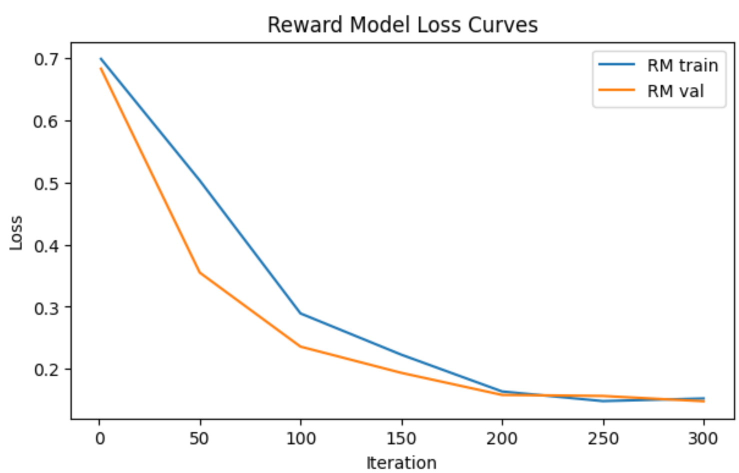
CS 639: Intro to Foundation Models
Homework 5

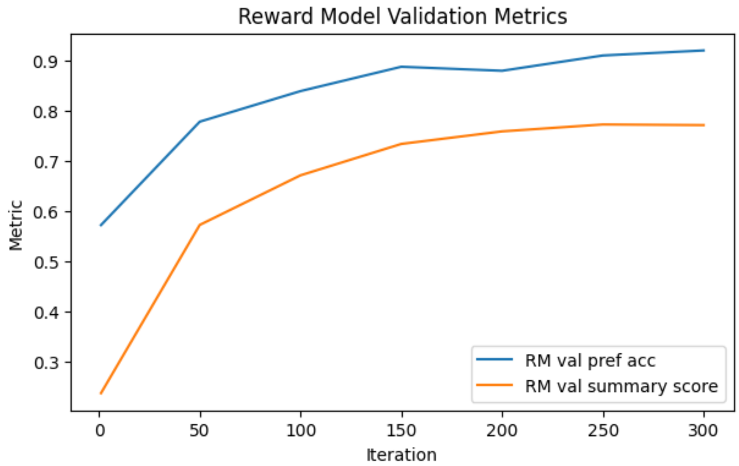
Amogh Dixit (asdixit3)

Part A: Core Alignment Implementation

(A7) The SFT and RM train/validation curves were obtained as below -





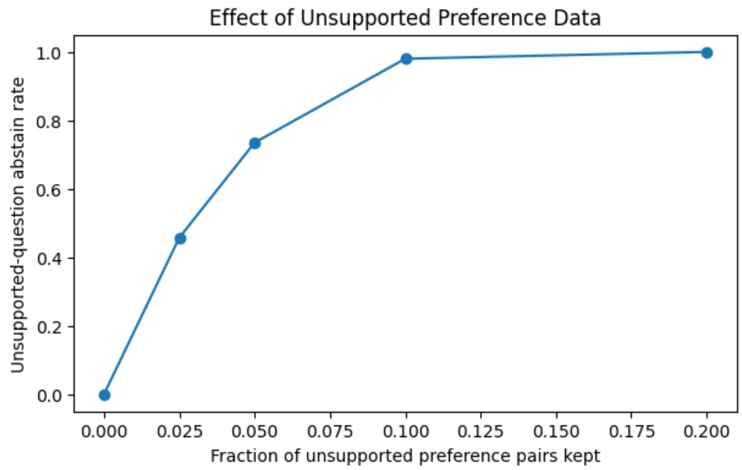


The SFT train and validation loss both decrease steadily over 1000 iterations. The final SFT metrics show perfect supported accuracy (1.00) but zero abstention on unsupported prompts. Since SFT only trains on factual responses, the model hallucinates on every unsupported query. After reward-model reranking, the behavior reverses, i.e. supported accuracy drops to 0.5425 while the unsupported abstain rate reaches 1.00, confirming the RM learned to prefer abstention.

Method	Supported Accuracy	Unsupported Abstain Rate	Hallucination Rate
SFT	1.0000	0.0000	1.0000
RM	0.5425	1.0000	0.0000

Part B: Alignment Experiments

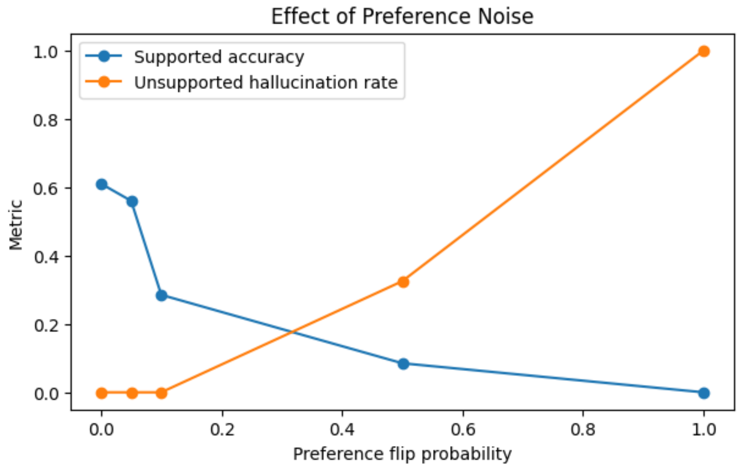
(B1) The unsupported-question abstain-rate plot was obtained as below -



When abstention is paired against every factual candidate as a negative, the reward model learns a global bias toward abstention rather than a context-sensitive rule. It cannot tell that abstention should only win when the fact is missing, it just sees abstention beating factual answers and so it applies that preference everywhere. This is why adding more unsupported data reduces supported accuracy. The model over-generalizes and starts preferring ”I don’t know” even when the context clearly contains the answer.

Fraction	Supported Accuracy	Unsupported Abstain Rate	Hallucination Rate
0%	0.9625	0.0000	1.0000
2.5%	0.7550	0.4575	0.5425
5%	0.3650	0.7350	0.2650
10%	0.4425	0.9800	0.0200
20%	0.3225	1.0000	0.0000

(B2) The combined metric plot was obtained as below -



Preference noise primarily hurts supported accuracy first, while unsupported abstention stays robust at low noise levels. This is because abstention is reinforced by more redundant pairs (four hallucinated negatives per prompt), so a few flipped labels don't change the overall signal much. Correct factual discrimination is a finer distinction and breaks down faster under noise. So noise doesn't degrade both behaviors equally. It widens the gap, making the model better at abstaining than at answering correctly.

Noise	Supported Acc	Unsupported Abstain Rate	Hallucination Rate
0%	0.610	1.000	0.000
5%	0.560	1.000	0.000
10%	0.285	1.000	0.000
50%	0.085	0.675	0.325
100%	0.000	0.000	1.000

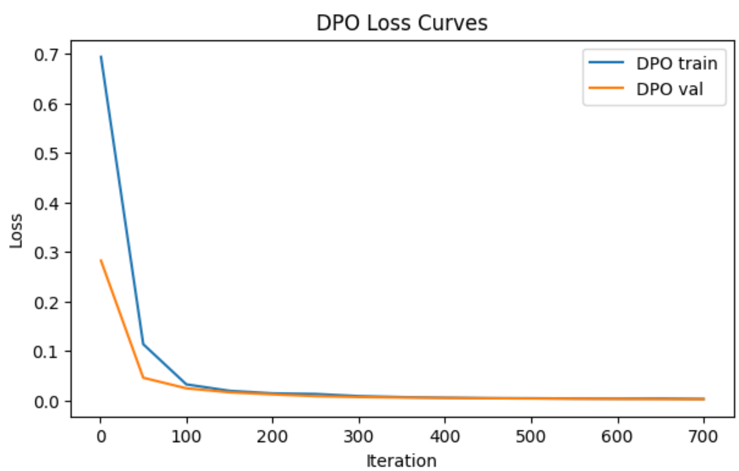
Part C: Direct Preference Optimization — Comparison and Failure Analysis

(C1)

Method	Supported Acc	Unsupported Abstain Rate	Hallucination Rate
SFT	1.0000	0.0000	1.0000
RM Reranking	0.5425	1.0000	0.0000
DPO	1.0000	0.0000	1.0000

Reward-model reranking works best here, achieving perfect abstention with zero hallucinations. DPO completely fails, matching the SFT baseline with 100% hallucination on unsupported prompts. RM reranking wins because it directly rescores each candidate at test time, so the preference signal immediately affects which response is picked. DPO instead tries to incorporate the preference into the policy's generation probabilities, but in a finite-candidate argmax setting the probabilities can shift in the right direction without ever flipping which candidate ranks first, so the observable behavior never changes.

(C2) The DPO loss curve was obtained as below -



The DPO loss decreases over 700 iterations, but the final behavior metrics did not change at all and are identical to the SFT baseline with supported accuracy = 1.00, unsupported abstain rate = 0.00, hallucination rate = 1.00. A decreasing DPO loss means the policy is increasing the log-probability margin $\log \pi_\theta(y_c \mid x) - \log \pi_\theta(y_r \mid x)$ in the right direction, but in a finite-candidate argmax setting this margin can grow without ever crossing the threshold that flips the top-ranked response from a factual answer to abstention. The SFT initialization strongly biases the policy toward factual answers, and 700 iterations are insufficient to overcome that inertia at decision time.